Enrollment No.: _____

Darshan UNIVERSITY

Darshan Institute of Engineering & Technology
B.Tech. | Sem-4 | Winter-2025

Course Code	:	2301CS401	**Date**	:	07-11-2025
Course Name	:	Database Management System - II	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1 **(A)** Define System Function and Under Defined Function. **4**

(B) Explain any three system functions with example. **3**

OR

Explain Stored Procedure with example.

(C) Explain cursor with its life cycle. **7**

OR

Explain cursor with its attributes.

Q.2 **(A)** Explain different Types of triggers. **4**

(B) Discuss transaction with the help of example. **3**

OR

Discuss Save Points in Transactions.

(C) Employees(EmpID, EmpName, Salary, DepartmentID) **7**
History(Description)
Create a trigger that insert a message 'An employee whose ID is XXX salary is updated and new salary is Rs. XXX' as new record in the History table whenever an employee's salary is updated.

OR

Employees(EmpID, EmpName, Salary, DepartmentID)
Departments(DepartmentID, DepartmentName, TotalSalary)
Create a trigger that updates the TotalSalary column in the Departments table whenever an employee's salary is updated.

Q.3 **(A)** Differentiate Clustered Index and Non-Clustered Index. **4**

(B) Explain types of execution plan. **3**

OR

Explain temporary table.

(C) What are the common warnings found in an execution plan? **7**

OR

What are the best practices for writing efficient SQL queries?

Q.4	**(A)**	Explain insertone and insertmany.	**4**
	(B)	Explain different types of NoSQL database.	**3**

OR

Explain advantages of NoSQL database.

	(C)	Discuss updateone and updatemany with example.	**7**

OR

Discuss deleteone and deletemany with example.

Q.5	**(A)**	Define Indexes in MongoDB with example.	**4**
	(B)	Explain BASE Theorem.	**3**

OR

Explain CAP Theorem.

	(C)	Explain Regex in MongoDB with example.	**7**

OR

Explain Aggregation in MongoDB with example.

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